

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	25 JULY 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

CAB BOOKING SYSTEM – TECHNICAL ARCHITECTURE

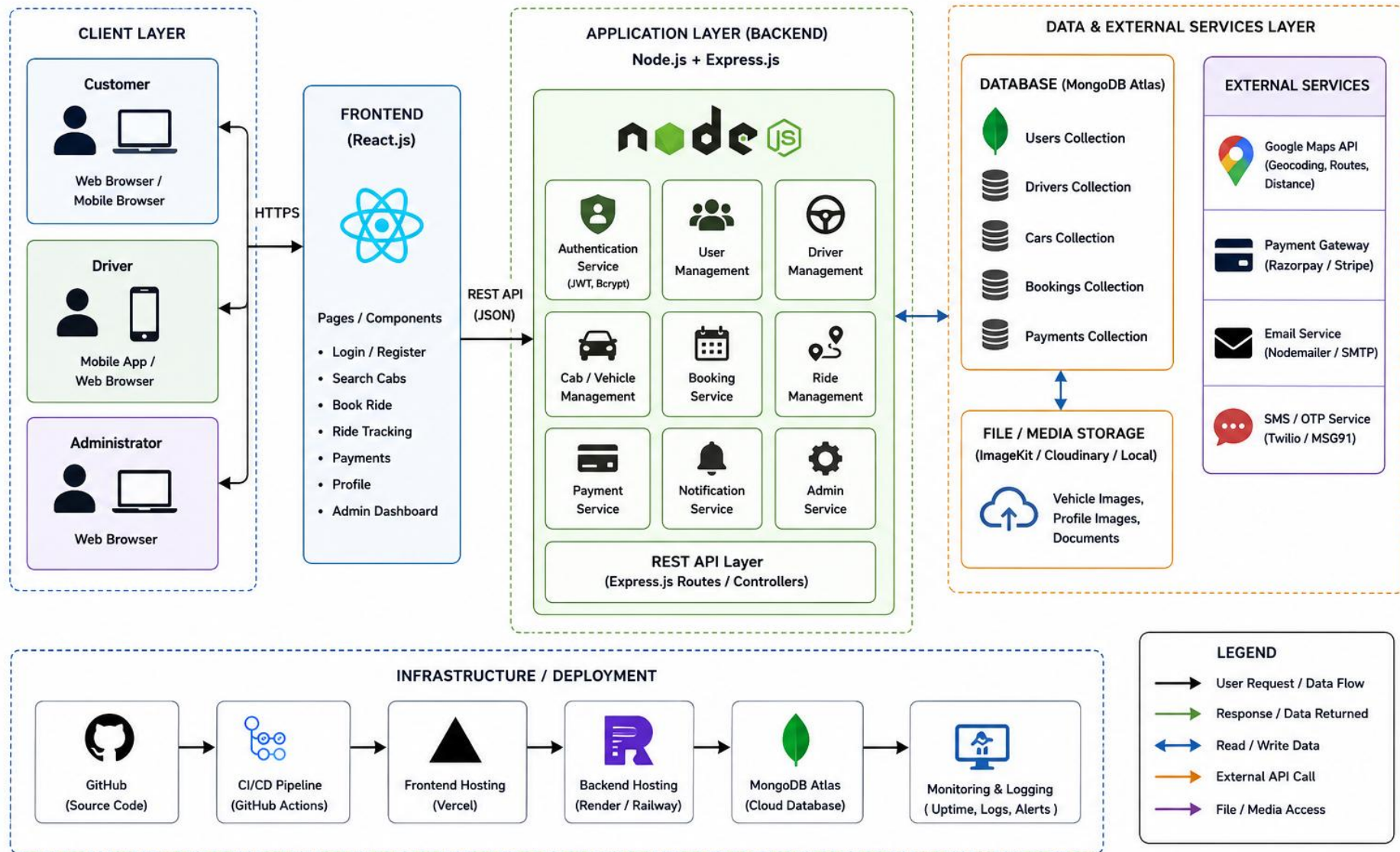


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web application for customers, drivers, and admin	React.js, HTML5, CSS3, JavaScript, Bootstrap/Tailwind CSS
2	Application Logic-1	User authentication and authorization	Node.js, Express.js, JWT, bcrypt
3	Application Logic-2	Cab booking, ride management, fare calculation	Node.js, Express.js
4	Application Logic-3	Payment processing and notifications	Express.js, Nodemailer, Payment API
5	Database	Stores users, drivers, cars, bookings, and payments	MongoDB, Mongoose
6	Cloud Database	Cloud-hosted database	MongoDB Atlas
7	File Storage	Stores vehicle images and user profile images	ImageKit / Local Storage
8	External API-1	Maps, route calculation, and location services	Google Maps API
9	External API-2	Secure online payment processing	Stripe / Razorpay API
10	Machine Learning Model	Not used in the current version	N/A
11	Infrastructure (Server/Cloud)	Application deployment	Vercel (Frontend), Render (Backend), MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend and backend development	React.js, Node.js, Express.js, MongoDB, Mongoose
2	Security Implementations	Secure authentication, password encryption, protected APIs, CORS	JWT, bcrypt, HTTPS, CORS, Environment Variables
3	Scalable Architecture	Three-tier architecture with separate frontend, backend, and database	MERN Stack, REST API
4	Availability	Cloud deployment with high availability and continuous access	Vercel, Render, MongoDB Atlas
5	Performance	Optimized database queries, image optimization, efficient REST APIs	MongoDB Indexing, Axios, ImageKit, Express Middleware